Z-C-1845
March 12, 1974
(See 6.5)

FEDERAL SPECIFICATION

CRABAPPLES, SPICED, CANNED

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers requirements for canned, spiced crabapples for use by all Federal agencies.

1.2 Classification.

1.2.1 Size and count. The spiced crabapples covered by this specification shall be of the following sizes and count per No. 10 can - (603 X 700) as specified (see 6.1).

Size	Count
Large	30 - 39
Medium	40 - 49
Small	50 - 60

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

TT-C-495 - Coatings, Exterior, for Tinned Food Cans
PPP-B-636 - Boxes, Shipping, Fiberboard
PPP-C-29 - Canned Subsistence Items, Packaging and Packing Of

Federal Standard:

FED-STD-123 - Marking for Domestic Shipment (Civil agencies)

FSC 8915

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(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-L-1497 - Labeling of Metal Cans for Subsistence Items
MIL-L-35078 - Loads, Unit; Preparation of Nonperishable Subsistence

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-668 - Sanitary Standards for Food Plants

(Copies of specifications, standards, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

PUBLICATIONS

Department of Health, Education and Welfare

Federal Food, Drug, and Cosmetic Act.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

United States Department of Argiculture

United States Standards for Condition of Food Containers

(Application for copies should be addressed to the Information Division, Agricultural Marketing Service, U.S. Department of Agriculture, Washington, DC 20250.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

Association of Official Analytical Chemists

Official Methods of Analysis of the Association of Official Analytical Chemists

(Application for copies should be addressed to the Association of Official Analytical Chemists, Post Office Box 540, Benjamin Franklin Station, Washington, DC 20044.)

National Motor Freight Traffic Association, Inc., Agent

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Material. All materials shall be clean, sound, wholesome and free from insect infestation or other objectionable matter, off flavors or off odors.

3.1.1 Crabapples. Crabapples shall be Hyslop or other suitable variety.

3.1.2 Sweetening ingredients. Sweetening ingredients shall be sugar (sucrose), liquid sugar, invert sugar or any combination thereof or a mixture of the aforementioned ingredients with dextrose or corn sirup, (liquid, dry or both) in quantities which do not adversely affect the sweetness nor the quality of the finished product.

3.1.3 Spices. Spices used shall be ground or whole and of full strength. Soluble spice flavoring of comparable strength may be used as an alternative for ground or whole spices.

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3.1.4 Food color. Coloring shall be US certified food colors.

3.1.5 Optional ingredient (vinegar). Cider vinegar, if used, shall be made by the alcoholic and subsequent acetous fermentation of the juice of apples. The finished product shall be clear, bright, brownish yellow in color, free from cloudiness, sediment, and floating particles, and shall have a clean, fruity odor.

3.2 Processing. The crabapples shall be washed, sorted and properly prepared for canning, The crabapples shall be packed in sirup solution to which food color has been added, and shall be processed to insure preservation and meet the requirements of 3.3.

3.3 Finished product.

3.3.1 Color, texture and flavor. The color shall be uniformly red, characteristic of artifically colored crabapples. The texture shall be firm but tender and not noticably mushy or mealy. The flavor shall be a good distinct sweet cinnamon-clove combination characteristic of properly prepared and processed spiced crabapples. The product shall be free of all off flavors.

3.3.2 Sizes. The crabapples shall be uniform in size not more than 7 percent by count shall be more than 5/16 inches smaller in diameter than the largest crabapple. The diameter of any crabapple shall be determined by measuring the diameter through the center, transverse to the longitudinal axis of the crabapple.

3.3.3 Sirup density and fill of container. The crabapples shall be covered or practically covered with the packing medium. The medium (sirup), 15 or more days after packaging shall be not less than 24° Brix at 20°C (68°F), when measured on a Brix hydrometer or by any other method giving equivalent results.

3.3.4 Defects. The product shall be practically free (not more than 5 percent by count) from scab spots which effect a total area of more than 1/4 inch in diameter or other serious blemishes e.g., those effecting appearance or eating quality, (not more than 5 percent by count of the crabapple shall be severely split). The product shall be practically free from harmless extraneous material, e.g., any vegetable substance including but not limited to leaves or portions thereof. There shall not be more than two pieces of harmless extraneous material per No. 10 (603 X 700) can. At least 50 percent by count of the crabapples shall have their stems attached.

3.4 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

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3.5 Workmanship. The product shall be prepared, processed and packaged under modern sanitary conditions and in accordance with good commercial practice. For military agencies, the product shall be processed in establishments meeting the requirements of MIL-STD-668.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling. Sampling of the finished product covered by this specification, and samples of any materials, components, or constituents entering into the preparation of the finished product may be taken at any time by the Government inspector to determine compliance with requirements.

4.2 Inspection.

4.2.1 General. Inspection shall be made by the procuring agency or a duly authorized representative at the time and place designated by the procuring agency. It may be made at the site of manufacture, both during and after processing and packing, at any suitable point in transit, or after delivery to the point of destination.

4.2.2 Final inspection. Unless otherwise specified (see 6.1), final inspection shall be made after delivery to the point of destination.

4.2.3 Condition inspection of containers. When inspection of product for quality and condition is performed by the U.S. Department of Agriculture, the United States Standards for Condition of Food Containers shall apply. (For applicable criteria governing inspection of containers for Military agencies see 4.3.3).

4.2.4 Responsibility for inspection. The supplier is responsible for meeting all the provisions of the specification prior to submission to the Government for acceptance. Unless otherwise specified in the contract or order, the supplier is required to perform the examination and tests prescribed herein, using his own or any other inspection facilities and services.

4.2.4.1 Inspection records. Inspection records of the examinations and tests shall be complete and made available for the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

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4.2.5 Preacceptance inspection (plant sanitation inspection).

4.2.5.1 Civil agencies. When specified (see 6.1), inspection of plants producing the product will be conducted upon request of the contracting officer to the appropriate Government inspection agency. Award of contracts will be limited to establishments known to maintain proper sanitary conditions or which have received prior sanitary approval recommended by the inspection agency.

4.2.5.2 Military agencies. The product shall be rejected if produced in plants not meeting the requirements of 3.5.

4.2.6 Sampling for inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except as otherwise indicated herein.

4.3 Component and material inspection. In accordance with 4.2.4. and 4.2.4.1, components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase documents.

4.3.1 Ingredients. Conformance of ingredients to identity (kind) requirements shall be ascertained by examination of labels, invoices or other valid documents. In addition, each ingredient to be used in the sirup shall be examined organoleptically to determine conformance to condition requirements of 3.1 and as may be contained in the pertinent ingredient paragraphs. The sample unit for testing shall be 8 ounces of ingredient derived from a primary container. The sample size shall be five. In the event a lot consists of less than five containers a sample unit shall be extracted from each container in the lot. Nonconformance with identity requirements or one or more sample units not conforming to condition requirements shall indicate an unacceptable ingredient and use of such shall be cause for rejection of the involved end item.

4.3.2 Process examination. Unless otherwise specified (see 6.1), examination shall be performed to the extent necessary to determine conformance to 3.2. Nonconformance shall be cause for rejection of the involved end item.

4.3.3 Examination of end item. Examination shall be in accordance with tables II through IV, and 4.3.3.1 through 4.3.3.3. The sampling plans for the aforementioned tables shall be in accordance with table I. The acceptable quality levels (AQL's) shall be expressed as percent defective for table II and defects per hundred units for tables III and IV.

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TABLE I. Sampling plans

Table	Inspection level	Sample unit	Lot size expressed in	AQLs Major	AQLs Minor	AQLs Totals
II	S-3	1 filled can	Cans	--	2.5	--
III	S-3	1 filled can	Cans	--	2.5	--
IV	S-2	1 filled can	Cans	1.5		6.5

TABLE II. Examination for drained weight and net (fill) weight 1/ 2/

Category	Defect
Minor	
201	Drained weight less than 54 ounces.
202	Net (fill) weight less than 104 ounces.

1/ See 4.4 for procedure, in determining drained weight.

2/ Report results to the nearest 1/2 ounce for net (fill) and drained weights. Lot shall be rejected if the average drained weight or net weight (fill) of the sample is less than 56 ounces or 104 ounces, as applicable.

Table III. Examination for count and size 1/

Category	Defect
Minor	
201	Count not within specified limits for large, medium or small, as applicable.
202	More than 7 percent (to nearest unit) by count, have a diameter more than 5/16 inch smaller than the largest apple.

1/ Measuring device shall be calibrated in 1/32 inch increments.

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TABLE IV. Examination of product 1/

Category		Defect
Major	Minor	
	201	Not a uniform red color.
	202	Less than 50 percent of apples by count have stems attached.
101		More than 5 percent (to nearest unit) of apples by count not free from scab or other serious blemishes, effecting total area of more than 1/4 inch in diameter.
102		More than 5 percent (to nearest unit) of apples by count severely split. 2/
103		More than 5 percent (to nearest unit) of apples by count mushy or mealy in texture.
	203	Three pieces of extraneous vegetative material.
104		More than three pieces of extraneous vegetative material.
105		Hard to the bite. 3/
106		Flavor not a good distinct sweet cinnamon-clove combination free of off flavors.
107		Mouth feel mealy. 3/

1/ Presence of foreign material (e.g. glass, paint, filth, insects, metal etc.) shall be cause for rejection of the lot.

2/ A severely split apple is one that has one or more splits in the flesh greater than 1/4 inches in width or one that has the skin split and curled back exposing more than 1/2 inch in width of flesh. Measuring device shall be calibrated in 1/32 inch increments.

3/ Examine one apple per sample unit.

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4.3.3.1 Examination of cans. Examination of cans for external condition and labeling shall be in accordance with the United States Standards for Condition of Food Containers. The examination of cans for interior coating and vacuum shall be in accordance with the applicable quality assurance provisions of PPP-C-29.

4.3.3.2 Examination of packing and shipping container. Examinations shall be in accordance with the quality assurance provisions of PPP-C-29.

4.3.3.3 Examination of unit loads. Unit loads shall be examined in accordance with the quality assurance provisions of MIL-L-35078.

4.3.4 Testing of finished product sirup. The sirups of the finished product shall be tested for soluble solids (Brix) not less than 15 days after the product is processed. Requirements and procedures shall be in accordance with 3.3.3 and 4.4.2.1. The sample unit for testing shall be 2-ounces of sirup derived from a can. The sample size shall be the number of cans indicated by inspection level S-2. The acceptable quality level expressed as percent defective shall be 1.5. Lot size shall be expressed in terms of cans. Results shall be reported to the nearest whole degree. A test failure shall be scored as a defect.

4.4 Examination and test methods. Examination and test procedures which differ from those specified herein, unless otherwise excepted, may be used by the supplier if they provide a quality assurance equivalent to that specified. If the Government contracting officer determines that such procedures and controls do not provide, as a minimum such quality assurance, the supplier will use the procedures set forth herein. In case of dispute as to examination and test results, the procedures specified herein will govern.

4.4.1 Examination procedure.

4.4.1.1 Examination for drained weight. The contents of the 603 x 700 can shall be poured into a 12 inch diameter U.S. Standard No. 8 (0.0937 inch square openings) sieve. Set sieve at 17 to 20° angle from horizontal, drain for two minutes and weigh. Drained weight is the crabapples and sieve less the weight of the sieve.

4.4.2 Test procedure.

4.4.2.1 Test procedures for determining Brix of finished product liquid media. The liquid media shall be tested after the product is canned, with a Brix hydrometer calibrated at 20°C. If the liquid media is tested at a temperature other than 20°C, the applicable temperature correction shall be made to the reading of the scale as prescribed in the Official Methods of Analysis of the Association of Official Analytical Chemists.

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5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be in accordance with level A or C as specified (see 6.1).

5.1.1 Level A. The product shall be packaged in a 603 x 700 can. The drained weight of the product shall be 56 ounces. A minus 2 ounce tolerance will be allowed in any one container, provided the average drained weight of the cans inspected in table II is not less than 56 ounces. The net (fill) weight per can shall be not less than 104 ounces. The can shall be an open-top style, round metal can with soldered side-seam and compound-lined double-seamed ends. The can shall be made throughout from commercial K-plate having a nominal tin coating weight of 0.75/0.25 pounds per base box or commercial differential electrolytic tin plate having a nominal plate weight of 1.00/0.25 pounds per base box. The entire inside area of the can (body and ends) shall be completely coated with enamel and the inside side-seam striped. The can materials (base plate, plating, interior coating, etc.) as specified above, shall be in accordance with the requirements in PPP-C-29. The cans shall be coated on the outside with a coating conforming to type I, or when specified, to type III of TT-C-495 (see 6.1). The hermetically sealed can shall show a vacuum of not less than 5 inches when examined in accordance with 4.3.3.1.

5.1.2 Level C. The product shall be packaged in accordance with 5.1.1 except that cans with or without commercial exterior coating will be acceptable.

5.2 Packing levels A, B and C. The product, packaged as specified in 5.1, shall be packed in accordance with level A, B or C of PPP-C-29, as specified (see 6.1).

5.2.1 When specified (see 6.1 and 6.3), the shipping container shall be reinforced with non-metallic strapping or pressure-sensitive adhesive filament reinforced tape in accordance with appendix of PPP-B-636.

5.3 Unit loads. When specified, the product packed as specified in 5.2 shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.2).

5.4 Labeling and marking.

5.4.1 Military agencies.

5.4.1.1 Cans. Cans shall be labeled in accordance with MIL-L-1497.

5.4.1.2 Shipping containers. Shipping containers shall be marked in accordance with MIL-STD-129.

5.4.1.3 Unit loads. Unit loads shall be marked in accordance with MIL-L-35078.

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5.4.2 Civil agencies.

5.4.2.1 Unit containers. Commercial labeling and additional labeling, as specified, that comply with the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder, is acceptable.

5.4.2.2 Shipping containers. Shipping containers shall be marked in accordance with the requirements of FED-STD-123 and such other information as specified by individual agencies.

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number and date of this specification.
(b) Count (size) required (see 1.2.1).
(c) When final inspection shall be made prior to delivery to point of destination (see 4.2.2).
(d) When plant sanitation inspection shall be performed by Civil agencies (see 4.2.5.1).
(e) When process examination is not required (see 4.3.2).
(f) Levels of packaging and packing required (see 5.1 and 5.2).
(g) When type III exterior coating of cans is required (see 5.1.1).
(h) When packing specified in 5.2.1 is required.
(i) Type and class of unit load, when unit loading is specified (see 5.3).

6.2 Estimated shelf life of the product, packed in tin cans, is approximately 18 months from the date of pack.

6.3 Packing specified in 5.2.1 is intended for transfer at sea operations or specific overseas operations.

6.4 Selection of levels (Military agencies). Based on the conditions known or expected to be encountered during shipment, handling and storage of the specific item being procured, the contracting officer should select the appropriate level of pack in accordance with the criteria established in AR 700-15, NAVSUPINST 4030.28, AFR 71-6, MCO 4030.14D or DSAR 4145.7, as applicable.

6.5 Supersession data. This specification includes the requirements of Interim Purchase Description S-1-0, dated 5 January 1970.

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Custodians:

Army - GL
Navy - SA
Air Force - 45

Review actiities:

Army - MD
Navy - MC, MS
DP - SS

Preparing activity:

Army - GL

CIVIL AGENCY COORDINATING ACTIVITIES:

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DOCUMENT IDENTIFIER AND TITLE
Crabapples, Spiced, Canned Z-C-1845

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

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